Norm Separation is Necessary: A Minimalist Calibration-Free INT3 Recipe for KV Cache Quantization

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Abstract

KV cache quantization is one of the principal levers for deploying long-context Transformer inference on fixed memory. We ask: at the INT3 tier (3 bits/value, 5.09× measured compression), is there a *minimalist* recipe (no calibration, no training, no codebook, roughly ten lines of code) that keeps WikiText-2 ΔPPL below 5 % of baseline on a broad set of open-weight Transformers?

We show that the composition of **norm separation, per-channel quantization, and an asymmetric `[min, max]` window** achieves this on eight open-weight Transformers spanning GPT-J (2021) through Qwen3.5-9B and Gemma-4-E4B (2026), reducing mean WikiText-2 ΔPPL by 66 % vs a symmetric per-channel variant at identical 3.0 bits/value.

A four-method ablation shows each ingredient is empirically necessary. Most informatively, the naive transplant of "asymmetric per-channel" from weight quantization (GPTQ, AWQ) to KV cache **fails on five of eight models, catastrophically on three**. A direct measurement localizes this failure to per-token L2 norm dispersion in the **V cache**: the Layer-0 V-CV predicts failure magnitude at Spearman $\rho \approx 0.88$, while the analogous K-CV does not ($\rho \approx 0.29$) — consistent with softmax partly absorbing K-side error but not V-side.

Two bounded downstream probes (MMLU 5-shot on two models and a 3-model multi-needle Needle-in-a-Haystack probe at 8 K and 32 K (pooled $n = 30$)) show no significant difference between the proposed recipe and FP16. Under the symmetric-only INT3 variant at 32 K, NIH pass rate drops to 50 % and produces cross-needle *bit-mixing hallucinations* that are absent under FP16 and under the proposed recipe.

All scripts and raw data are released under Apache 2.0. Scope limitations: downstream coverage is two probes on limited model subsets, throughput is not measured, and the Qwen2-7B INT4 valley is an $n = 1$ observation. Further detail on the Gemma-4-E4B under stricter protocols (§5.1 / §7), the confounder caveat on the V-CV mechanism (§5.3), and the 8 K Qwen2-7B NIH anomaly (§5.7) lives in the body.

Contributions

Concretely, this paper contributes:

1. **A calibration-free INT3 recipe** (`qa_nsep_pchan_asym`) that composes norm separation, per-channel scaling, and an asymmetric quantization window. The reference implementation is a ten-line Python function with no external data, optimizer state, or learned parameters (§3).

2. **An eight-model end-to-end validation on WikiText-2 sliding-window perplexity** (ctx = 1024, 10 windows) spanning 2021–2026: GPT-J-6B, Pythia-6.9B, Mistral-7B-v0.1, Qwen2-7B, Qwen2.5-14B, Llama-3.1-8B, Qwen3.5-9B, Gemma-4-E4B. Under this primary protocol, all eight stay below our 5 %-of-baseline-PPL threshold; the mean ΔPPL reduction over a symmetric `nsep+pchan` baseline is 66 % at identical 3.0 bits/value (§5.1). A follow-up bf16 / 30-window probe places Gemma-4-E4B at 5.5 %–7.0 % of baseline, so that specific model is borderline rather than uniformly below threshold under stricter protocols.

3. **A four-method ablation establishing that each of the three components is empirically necessary.** No simpler ablation variant ((A) naive per-row symmetric, (B) asymmetric per-channel without norm separation, or (C) symmetric nsep+pchan) achieves the eight-model coverage of the full recipe (D) (§5.2).

4. **A specific, reproducible negative result for a naive translation of weight-quantization recipes to KV caches, with a direct measurement of the mechanism.** Variant (B) above (the standard "asymmetric per-channel" baseline as it appears in GPTQ/AWQ-style weight quantization) **fails our usability threshold on five of the eight models: catastrophically on three (Llama-3.1-8B, Mistral-7B-v0.1, Qwen2.5-14B, with regression up to 24× vs naive per-row INT3 on Llama-3.1-8B) and mildly on GPT-J-6B (6.27 %) and Pythia-6.9B (5.88 %). It is also actively worse than the naive per-row baseline (A) on four of the eight (Mistral, Qwen2.5-14B, Llama-3.1-8B, GPT-J-6B).** A direct measurement of per-token L2 norm dispersion on Layer-0 K and V caches across all eight models (§5.3, Appendix E) supports a refined mechanism claim: **L0 V-cache per-token L2 norm CV correlates strongly with (B)'s regression magnitude (Spearman $\rho \approx 0.88$), while L0 K-cache CV does not ($\rho \approx 0.29$)**. We therefore argue that `nsep` is a structural ingredient for per-channel KV quantization specifically because V-cache reconstruction error bypasses the softmax normalization that partially absorbs K-cache error, rather than a generic quality-of-life preprocessing step (§5.3).

5. **Documented negative results on rotation and top-K mixed precision**, both tested during the development of this recipe. A Walsh–Hadamard rotation step (similar in spirit to KVLinC and the rotation component of TurboQuant) regresses per-head reconstruction MSE by 3.68× on real Qwen2-7B Layer 0 K; a top-K outlier-channel mixed-precision variant at $K = 4$ regresses per-head reconstruction MSE by 17.6 % on the same tensor. Both metrics are per-tensor reconstruction error rather than end-to-end ΔPPL; we did not pursue either method far enough to run the full PPL pipeline. Both quantizer implementations are retained in the public code base as documented negative results (§6, Appendix A).

6. **An open observation on accidental K-cache entropy regularization.** On Gemma-4-E4B evaluated text-only, variant (B) produces a small but reproducible *negative* ΔPPL (−0.32 to −0.44 on two non-overlapping WikiText-2 segments). A K-only vs V-only split attributes 87 % of the effect to the Key cache, consistent with quantization noise softening an over-confident attention distribution on an off-distribution (text-only) input to a multimodal-trained model. We do not promote this as a method (the same ablation fails the threshold on five other models (catastrophically on three, mildly on two)) but record it as a phenomenon worth instrumenting in future work (§7).

7. **An MMLU 5-shot downstream probe showing no detectable accuracy cost on the two canonical benchmark models.** At 500 test questions across five subjects, asymmetric INT3 (the v0.3 recipe) delivers accuracy deltas of +1.00 pp on Qwen2-7B and −0.20 pp on Llama-3.1-8B relative to the FP16 baseline — both within the ≈ 3 pp conservative independent-proportion delta standard error at this sample size (§5.6). This is a single-task probe and does not replace a full downstream campaign; we discuss scope in §5.6.

8. **A 3-model long-context multi-needle retrieval probe where the v0.3 recipe shows no significant degradation vs FP16 while the v0.2 symmetric recipe does.** A distractor-hardened Needle-in-a-Haystack protocol scored on lead-key retrieval (Llama-3.1-8B, Qwen2-7B, and Qwen2.5-14B, 10 trials per cell, pooled $n = 30$) shows that at 32 K tokens the asymmetric recipe (D) retains 83.3 % pass rate with **no significant difference vs the 96.7 % FP16 baseline under this exploratory test** (Fisher exact two-sided $p = 0.194$; $n = 30$ is too small for a formal non-inferiority claim), while the symmetric recipe (C) drops to 50 % (FP16-vs-C $p = 6.4 \times 10^{-5}$; C-vs-D $p = 0.013$) (§5.7). Under (C) at 32 K, all three models produce cross-needle **bit-mixing hallucinations** (e.g., `"XRZ5-BLUE"` on Llama, `"LMQ3-GREEN"` on Qwen2-7B, `"WZQ7-BLUE"` on Qwen2.5-14B); (D) produces zero such hallucinations across 30 asymmetric-INT3 trials at 32 K. One open observation (Qwen2-7B (Mode 1) at 8 K context shows a within-needle digit-corruption pattern under (D)) is flagged as a Mode-1-specific short-context interaction to study in future work.

We also argue, as a secondary contribution, that the **"Mode 1" phenomenon** (a non-monotonic INT4 ΔPPL valley observed on Qwen2-7B (n = 1 in our sample)) is a concrete, reproducible failure case of naive KV quantization that motivates looking for robust recipes in the first place. We emphasize throughout that this remains an n = 1 observation and explicitly do not claim it generalizes to untested models (§2, §5.3).

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1. Introduction

Transformer inference at long context length is increasingly memory-bound on the KV cache rather than on weights. For a Pre-LN decoder with L layers, H key/value heads, head dimension D, sequence length T, and FP16 storage, the KV cache occupies $2 \cdot L \cdot H \cdot T \cdot D \cdot 2$ bytes per sequence. At the 128 K and 1 M context lengths now common in open-weight models (Llama-3.1, Qwen2.5, Qwen3.5, Gemma-4), the cache dominates per-sequence GPU memory by an order of magnitude over the model weights themselves. Quantizing the KV cache has therefore become one of the principal levers for practical deployment, and a body of recent work has pushed effective bit rates from the FP16 baseline toward 4-bit, 3-bit, and in some settings 2-bit or 1.5-bit representations.

The current state of KV cache quantization is, however, methodologically heterogeneous. **KIVI** [Liu et al., 2024] uses asymmetric per-channel for keys and per-token for values at 2 bits and introduces outlier protection logic. **KVQuant** [Hooper et al., 2024] uses per-channel key quantization in the pre-RoPE frame with a calibration-based non-uniform quantizer. **PolarQuant** [Wang et al., 2025a] replaces the Euclidean coordinates of each KV vector with a polar-coordinate decomposition and trains a k-means codebook on the polar angles after a random orthogonal rotation. **TurboQuant** [Google Research, 2026] applies a random orthogonal rotation, a Lloyd–Max scalar quantizer on the rotated coordinates, and a QJL-style residual correction to obtain a data-oblivious recipe that is competitive at the 3-bit tier. **KVLinC** [KVLinC authors, 2025] combines a Hadamard rotation with learned linear correction adapters to push 2-bit KV cache on Llama and Qwen families. Each of these methods delivers stronger compression ratios or finer-grained precision than a naive scalar quantizer, but each introduces a corresponding complexity dependency: a codebook, a calibration corpus, a rotation matrix (random or Hadamard), a trained adapter, or a non-trivial mixed-precision scheme.

Against this backdrop we consider a narrower and more practical question. **Is there a recipe that stays at the INT3 tier** (3.0 bits/value, approximately $16/3 = 5.33\times$ theoretical compression, $\sim 5.09\times$ measured with FP16 metadata overhead) **that requires no calibration, no codebook, no rotation, no adapter, fits in roughly ten lines of reference code, and still keeps the WikiText-2 perplexity drift below a conservative quality threshold on a broad slice of modern Transformers**? Each individual ingredient of such a recipe is, separately, folklore: norm separation ($x = n \cdot d$, with n stored in FP precision) is the structural idea underlying PolarQuant and TurboQuant; per-channel absmax scaling is the standard per-channel preconditioner used throughout weight quantization since LLM.int8 [Dettmers et al., 2022]; asymmetric [min, max] quantization windows are the default in GPTQ [Frantar et al., 2023] and AWQ [Lin et al., 2024] for *weights*. The question of interest is not the novelty of any ingredient but the empirical behavior of a particular composition of them on *KV caches* across a representative model population.

We study the composition `nsep+pchan+asym` (per-channel asymmetric INT3 quantization applied to the unit-direction vector after norm separation) and report three main findings. First, on an eight-model benchmark spanning 2021 (GPT-J-6B) through 2026 (Qwen3.5-9B, Gemma-4-E4B), this recipe keeps the WikiText-2 sliding-window ΔPPL below 5 % of the FP16 baseline on every model we were able to run under our primary protocol (fp16, 10 windows × 1024 ctx), at exact 3.0 bits/value with measured 5.09× compression (§5.1, §5.4). Second, in a four-method ablation on the same eight models (§5.2), none of the three simpler variants (naive per-row symmetric, asymmetric per-channel *without* norm separation, symmetric nsep+pchan) achieves the same coverage; each variant falls outside the 5 % threshold on at least two models, and the asymmetric-without-nsep variant in particular regresses catastrophically (> 100 % of baseline) on three of them. Third, the most informative ablation failure is the naive asymmetric-per-channel variant that omits norm separation: this recipe, which is approximately what one obtains by copy-pasting the asymmetric per-channel machinery of AWQ/GPTQ from weights to KV cache, *fails the threshold on 5 of the 8 models* (and is worse than the naive per-row baseline on 4 of them), reaching +10.92 ΔPPL on Llama-3.1-8B — a 24× regression relative to naive per-row INT3 on the same tensor (§5.3). We trace this failure to a specific structural property of KV caches (V-cache per-token L2 norm dispersion, which correlates with the failure magnitude at Spearman $\rho \approx 0.88$ while the corresponding K-cache statistic does not ($\rho \approx 0.29$)) that weight tensors largely lack, and we argue that norm separation therefore is not a quality-of-life preprocessing step but a structural requirement for per-channel KV quantization on this model family.

The benchmark also includes one anomaly that we treat explicitly as $n = 1$: Qwen2-7B exhibits a non-monotonic INT4 failure under naive per-row quantization (+1382 ΔPPL at INT4 versus +12 at INT3 and +5 at INT5), which we trace to an outlier channel at Layer 0 of the K cache. This "INT4 valley" is the one observation of this pattern in our 27-model broader survey (seven of which we re-benchmarked end-to-end for the present work). We report it as a concrete failure mode that motivates looking for recipes robust across the bit-width axis, and we are explicit throughout that it is a single-model observation; we do not claim a general "Mode 1" class of models exists. The proposed recipe handles this anomaly alongside the common case at identical bit budget (§5.1, §6). Finally, we document one curious *beneficial* anomaly — a reproducible small negative ΔPPL on Gemma-4-E4B under the asymmetric-without-nsep variant, traced at 87 % to the Key cache and plausibly caused by accidental attention-logit entropy regularization on a multimodal-trained model evaluated text-only (§7). We include this only as an observation in search of a theory; it is not a deployment recommendation.

The remainder of the paper is organized as follows. §2 surveys related work. §3 specifies the `nsep+pchan+asym` recipe in closed form. §4 describes our experimental setup. §5 reports the main results, including the four-method ablation. §6 discusses limitations — in particular our reliance on a single-benchmark (WikiText-2) proxy for quality, the $n = 1$ status of the Qwen2-7B anomaly, and Python-implementation overhead that our measurements do not attempt to remove. §7 documents the Gemma-4-E4B K-cache regularization observation. §8 concludes. Reproducibility materials, including dead-end experiments, the full Phase 13 method sweep, and the complete set of measured JSON results, are provided in the appendices and in the public code repository.

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2. Related Work

KV cache quantization divides roughly into three families, plus a fourth category of ingredients we transplant from weight / activation quantization.

Per-channel / per-token scalar quantization with outlier handling. KIVI [Liu et al., 2024] observes that keys and values have different distribution structure (keys benefit from per-channel quantization while values benefit from per-token) and implements aggressive 2-bit quantization with explicit outlier reservation. KVQuant [Hooper et al., 2024] performs per-channel quantization of keys in the pre-RoPE frame, where the distribution is cleaner, and uses calibration data to fit a non-uniform quantizer. Atom [Zhao et al., 2024] combines group-wise mixed-precision quantization with explicit outlier isolation on both K and V, reaching 4-bit end-to-end serving quality with per-channel statistics plus a dedicated high-precision slot for outlier channels. All three methods rely on some combination of calibration, explicit outlier reservation, or group-wise bit allocation; the recipe we propose is simpler (no outlier slot, no calibration) at a comparable or slightly higher bit rate.

Polar / norm-direction decomposition with codebooks. PolarQuant [Wang et al., 2025a] decomposes each KV vector into its L2 radius (stored at full precision) and its angular coordinates on the unit sphere, the latter quantized via a k-means codebook fit after a random orthogonal rotation. TurboQuant [Google Research, 2026] builds on a similar random-rotation idea by applying a Lloyd-Max scalar quantizer on the rotated coordinates with a QJL-style residual correction, yielding a fully data-oblivious recipe. Our work shares the norm-direction decomposition with these methods but deliberately uses the simplest possible direction quantizer (per-channel absmax with asymmetric window) instead of a codebook or a rotation-plus-residual scheme.

Rotation-based quantization. KVLinC [KVLinC authors, 2025] applies a Hadamard rotation to the K cache and trains small linear-correction adapters that compensate for the residual quantization error. Outside the KV-cache-specific setting, QuaRot [Ashkboos et al., 2024] applies a fixed Hadamard rotation to weights and activations to enable 4-bit inference, and SpinQuant [Liu et al., 2024] learns the rotation matrix end-to-end; both target weight+activation quantization rather than KV caches per se. Rotation spreads outlier energy across coordinates and is broadly effective at 2 bits in the KVLinC pipeline. In early development of the present work we tested a fixed Walsh–Hadamard rotation as a drop-in preprocessing step before `nsep+pchan` and observed a $3.68\times$ per-head reconstruction-MSE regression on real Qwen2-7B Layer 0 K, which we attribute to the specific multi-channel plateau structure of that tensor (§A.1). We therefore do not rotate in the proposed recipe; the present method obtains its generality from norm separation plus per-channel asymmetric scaling alone, without learned correction. This is a negative result for *fixed Walsh–Hadamard rotation pre-pended to* `nsep+pchan`; it is not a negative result for the KVLinC pipeline in its published form, which combines rotation with an adapter and targets 2-bit rather than 3-bit regimes.

Subspace-orthogonal projection. Beyond rotation proper, SQuat [Wang et al., 2025b] is a calibration-free K-cache quantizer that builds a subspace from the query tensors and constrains the (de)quantization error on keys to lie in the *orthogonal complement* of that subspace — engineering the K-side residual to land in directions attention discards. This is methodologically complementary to our §5.3 finding that Layer-0 K-cache per-token norm CV does *not* predict variant-(B) failure (Spearman $\rho \approx 0.29$) while the Layer-0 V-cache CV does ($\rho \approx 0.88$): where SQuat actively steers K error into attention-irrelevant directions, we observe that K-side error is already partly absorbed by softmax normalization and is therefore the smaller of the two concerns at INT3. The two findings point in the same direction (K-side error is cheaper to tolerate than V-side error) from opposite sides of the attention operator.

Weight- and activation-quantization ingredients transplanted to KV. Per-channel scaling as a preconditioner for low-bit Transformer quantization traces to LLM.int8 [Dettmers et al., 2022] for outlier isolation and to SmoothQuant [Xiao et al., 2023], which migrates the activation dynamic range onto the weights so that per-channel activation quantization becomes feasible at INT8. Asymmetric per-channel $[\min, \max]$ quantization is the standard preconditioner in GPTQ [Frantar et al., 2023], AWQ [Lin et al., 2024], and SpQR [Dettmers et al., 2024] on the weight side. To our knowledge there is no work that explicitly isolates whether this ingredient transplants cleanly from weights or activations to KV caches when applied naively. §5.3 of this paper reports that it does *not*: variant (B) in our ablation, which is close to the naive transplant, fails our threshold on five of the eight models we test (three catastrophically, two mildly) and is worse than the naive per-row baseline on four of them. A per-model diagnostic (§5.3, Appendix E) localizes the root cause to V-cache per-token L2 norm variance (large on older architectures, bounded mechanically on 2026 QK-normalized models) which contaminates per-channel statistics when norm separation is omitted and then propagates linearly through the attention-weighted output; the corresponding K-cache variance is partly absorbed by softmax normalization and does not predict the failure.

Positioning. We view this paper as complementary to the above work rather than competing at the bit-rate frontier. For *sub-INT3 / 2-bit* compression, KVLinC's Hadamard rotation plus learned adapters, PolarQuant's polar-angle codebook, TurboQuant's rotated-Lloyd-Max-plus-QJL pipeline, and KIVI's asymmetric 2-bit per-channel/per-token scheme each push the compression ratio past what a calibration-free scalar quantizer can deliver. For *outlier-aware 4-bit serving*, Atom's mixed-precision-plus-outlier-slot design targets a different operating point: strong per-token throughput with an explicit protected slot for outlier channels. We offer a minimalist recipe at the INT3 tier whose contribution is a different property set (*generality-without-calibration* and the empirically-supported minimality of its three ingredients) and do not compete with either regime head-on. Where a sub-2-bit budget is the constraint, KVLinC or a PolarQuant/TurboQuant-style codebook is likely the better choice; where 4-bit outlier-aware serving is the constraint, Atom is the better fit; where a calibration-free, codebook-free, adapter-free drop-in is preferred and 3 bits/value is acceptable, the recipe here has a different set of properties.

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3. Method

3.1 Setup and notation

Consider a single key or value tensor from one head of a Transformer's KV cache, shaped $[T, D]$, where T is the number of cached tokens and D is the head dimension. Let $x_t \in \mathbb{R}^D$ denote the row corresponding to token t. We evaluate quantizers that map x_t to a low-precision representation $\hat{x}_t$ and measure the end-to-end effect via WikiText-2 perplexity on a Transformer whose KV cache has been replaced by the reconstruction. Throughout, *bits per value* refers to the storage cost in the reconstructed tensor averaged over coordinates — including any per-channel or per-token metadata that is stored alongside the packed integer data.

Scope of the "minimal" claim. Where we later describe the proposed recipe as "minimalist", we mean it within the class of *scalar, data-independent* quantizers — schemes that compute all parameters directly from the tensor statistics without a calibration pass, a learned adapter, a non-uniform codebook, or a rotation matrix. We do not claim minimality against learned quantizers or calibrated codebooks, which can be simpler in other senses (e.g., lower bit rate for the same quality); the trade we study is a different operating point in which the recipe itself carries no state outside the per-channel and per-token scalars it computes from the input.

3.2 The recipe

The method has three data-independent operations applied in sequence on every $[T, D]$ key or value tensor.

Step 1 — Norm separation (`nsep`). Decompose each row into magnitude and direction:

$$n_t = \|x_t\|_2, \qquad d_t = \frac{x_t}{n_t}.$$

We store $n_t \in$ FP16 exactly; only the unit direction d_t is quantized.

Step 2 — Per-channel asymmetric window (`pchan` + `asym`). For each channel $c \in \{1, \dots, D\}$, compute

$$m_c = \min_t d_{t,c}, \qquad M_c = \max_t d_{t,c}, \qquad s_c = \frac{M_c - m_c}{2^b - 1},$$

where b is the target bit width ($b = 3$ in our main results). We store $m_c, s_c \in$ FP16.

Step 3 — Integer quantization + re-normalization. Quantize each direction coordinate to a b-bit unsigned index:

$$q_{t,c} = \text{round}\left(\frac{d_{t,c} - m_c}{s_c}\right), \qquad q_{t,c} \in \{0, 1, \dots, 2^b - 1\}.$$

At reconstruction time, the dequantized direction is

$$\tilde{d}_{t,c} = m_c + s_c \cdot q_{t,c},$$

and we re-normalize onto the unit sphere before multiplying by the stored norm:

$$\hat{d}_t = \frac{\tilde{d}_t}{\|\tilde{d}_t\|_2 + \varepsilon}, \qquad \hat{x}_t = n_t \cdot \hat{d}_t.$$

The re-normalization step (cheap, one FP32 norm per token) corrects the small radial drift that per-coordinate quantization would otherwise introduce on the unit sphere.

The entire reconstruction pipeline is $O(TD)$ time per tensor, requires no calibration corpus, no learned parameters, and no transformation that is tensor-dependent beyond the two per-channel scalars (m_c, s_c).

3.3 Storage accounting

Per key or value tensor of shape $[T, D]$, the recipe stores: - $T \cdot D$ integers of b bits each (packed); - T FP16 norms (one n_t per token); - $2D$ FP16 per-channel scalars (m_c and s_c).

The average bit rate per coordinate is therefore

$$b_{\text{eff}}^{\text{asym}}(b, T, D) \;=\; b \;+\; \underbrace{\frac{16}{D}}_{\text{per-token norm}} \;+\; \underbrace{\frac{32}{T}}_{\text{two per-channel scalars}}.$$

At $b = 3$, $D = 128$, $T = 2048$ (representative of the main results), $b_{\text{eff}}^{\text{asym}} = 3 + 0.125 + 0.0156 \approx 3.141$ bits/value. At $T \to \infty$ this asymptotes to $3 + 16/D = 3.125$ bits/value.

For comparison, the symmetric per-channel variant used as the §5 baseline stores only one FP16 scalar per channel (the absmax) instead of two, so its bit rate is

$$b_{\text{eff}}^{\text{sym}}(b, T, D) \;=\; b \;+\; \frac{16}{D} \;+\; \frac{16}{T}.$$

The asymmetric variant's overhead relative to the symmetric one is therefore exactly $b_{\text{eff}}^{\text{asym}} - b_{\text{eff}}^{\text{sym}} = 16/T$ — that is, one additional FP16 per channel, amortized over the T tokens that share that channel's scalar. At $T = 2048$ this is ≈ 0.0078 bits/value (matching the Phase 15 measurement in §5.4 of $3.141 - 3.133 = 0.008$ bits/value). The extra storage cost is negligible at any realistic context length; the main quality trade is not paid in bits.

3.4 Real-packed storage

In the reference implementation, three-bit integers are packed using the existing `pack_int3_unsigned` primitive in the modeq code base: every 8 consecutive 3-bit values fit into 3 bytes. The packed representation is bit-exact: pack-then-unpack is the identity on integers in $\{0, \dots, 7\}$. On Qwen2-7B (28 layers × 4 KV heads × 2048 tokens × head dim 128) and Llama-3.1-8B ($32 \times 8 \times 2048 \times 128$), the measured on-GPU cache size drops from 112 MB → 21.98 MB (Qwen2-7B) and 256 MB → 50.25 MB (Llama-3.1-8B), both at **5.09× compression vs FP16** (§5.4). The symmetric variant achieves 5.11×; the asymmetric variant's $0.02\times$ deficit is purely the extra per-channel scalar mentioned in §3.3 and is the entire price paid for the 58–95 % ΔPPL reductions reported in §5.

3.5 Reference implementation

```python
def qa_nsep_pchan_asym(x, bits=3):
    """Norm-separated per-channel asymmetric INT quantization.

    x: shape [..., T, D], dtype fp16 or fp32.
    Returns a same-shape, same-dtype tensor reconstructed from the
    INT-quantized direction vector plus the exact per-token L2 norm.
    """
    dtype = x.dtype
    xf = x.float()
    n = xf.norm(dim=-1, keepdim=True).clamp(min=1e-12)  # per-token
    d = xf / n                                          # unit direction
    m = d.amin(dim=-2, keepdim=True)                    # per-channel
    M = d.amax(dim=-2, keepdim=True)
    step = ((M - m) / (2**bits - 1)).clamp(min=1e-12)
    q = ((d - m) / step).round().clamp(0, 2**bits - 1)  # integer index
    d_q = m + q * step                                  # dequantize
    d_q = d_q / d_q.norm(dim=-1, keepdim=True).clamp(min=1e-8)
    return (n * d_q).to(dtype)
```

The function has ten lines of content plus docstring; it performs no I/O, has no state, and is independent of the caller's model. A matching packed-cache class (`Int3AsymNsepPchanCache`) handles real bit packing, exactly reproduces the reconstruction above, and exposes a `size_bytes()` method.

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4. Experimental Setup

Benchmark. All ΔPPL measurements are on WikiText-2 [Merity et al., 2017] (test split) under a sliding-window protocol: context length $= 1024$ tokens, stride $= 512$, and up to 10 windows per evaluation unless otherwise stated. Each window is split in half: the first 512 tokens are fed without target loss to populate the KV cache; the KV cache is then optionally replaced by a quantized reconstruction; the second 512 tokens are scored. This matches the sliding-window perplexity protocol used by PolarQuant and KIVI. Baseline PPL is the model's own FP16 PPL under the same protocol with no quantization.

Quality metric. We report $\Delta\text{PPL} = \text{PPL}(\text{quantized}) - \text{PPL}(\text{baseline})$ in absolute units and as a fraction of the baseline (denoted "% of baseline PPL"). We adopt a fixed 5 % threshold as a conservative proxy for "likely usable in practice if memory is the dominant constraint" and call it the *usability threshold* throughout. This threshold is modeq-internal; it is not a deployment certification. §5.6 reports one exploratory MMLU 5-shot downstream probe on two models as a supporting data point, but broader downstream-task coverage (reasoning, code, long context, instruction following) remains future work.

Models. Eight open-weight Transformers spanning 2021–2026:

Model	Release	D	H_{KV}	Family
GPT-J-6B	2021	256	16	pre-Llama-family
Pythia-6.9B	2023	128	32	Pythia
Mistral-7B-v0.1	2023	128	8	Mistral
Qwen2-7B	2024	128	4	Qwen / GQA
Qwen2.5-14B	2024	128	8	Qwen / GQA
Llama-3.1-8B	2024	128	8	Llama-3-family
Qwen3.5-9B	2026-02	256	4	Qwen / hybrid
Gemma-4-E4B	2026-04	256	4	Gemma-4 Efficient

Qwen3.5-9B uses a hybrid attention architecture in which 8 of 32 layers are standard attention (with K/V cache) and the remaining 24 are linear attention (no K/V cache); the compression pipeline skips non-KV layers and compresses only the 8 standard-attention layers. Gemma-4-E4B is multimodal-capable but is evaluated text-only in all measurements below. We attempted a ninth model, Phi-3-medium-4k-instruct, but its bundled custom `modeling_phi3.py` is incompatible with current `transformers` on three separate fronts (rope-scaling key rename, unknown `rope_type = "default"`, removed `DynamicCache.from_legacy_cache`); we count it as "attempted but blocked" rather than as a failure of the recipe.

Hardware and reproducibility. All measurements were collected on a single NVIDIA A100 (40 GB) in Google Colab. The recipe is implemented in the `modeq` open-source package; every ΔPPL number and compression ratio reported below is reproducible from the committed JSON data under `results/` via a matching script under `experiments/`. The full set of quantizer primitives fits in under 50 lines of Python; no custom CUDA kernel is used and we do not claim any throughput advantage over the FP16 baseline (§6).

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5. Results

5.1. Eight-model generalization at INT3

Table 1 reports the WikiText-2 ΔPPL under our recipe (the `nsep+pchan+asym` variant at $b = 3$) against the v0.2 symmetric baseline at the same bit budget. Both methods operate at exactly 3.0 bits/value plus $O(D/T)$ FP16 metadata, so the comparison is at matched bit rate.

Table 1. Eight-model WikiText-2 ΔPPL at INT3 (ctx = 1024, 10 windows). "sym" = symmetric nsep+pchan (this paper's v0.2 baseline variant). "asym" = the recipe proposed in §3. All numbers below use the primary fp16 / 10-window protocol; Gemma-4-E4B under a follow-up bf16 / 30-window probe lands at 5.5 %–7.0 % of baseline depending on text segment (§7, Appendix C) and is therefore borderline rather than uniformly below threshold under stricter protocols.

Model	base PPL	sym INT3 ΔPPL	asym INT3 ΔPPL	reduction	% of base
GPT-J-6B	10.29	+0.1873	**+0.0703**	−62.5 %	0.68 %
Pythia-6.9B	10.76	+2.3781	**+0.0994**	−95.8 %	0.92 %
Mistral-7B-v0.1	6.30	+0.1243	**+0.0547**	−56.0 %	0.87 %
Qwen2-7B	8.37	+0.4021	**+0.1034**	−74.3 %	1.24 %
Qwen2.5-14B	1.96	+0.2608	**+0.0909**	−65.1 %	4.64 %
Llama-3.1-8B	7.38	+0.2541	**+0.1059**	−58.3 %	1.43 %
Qwen3.5-9B	10.32	+0.2750	**+0.0942**	−65.7 %	0.91 %
Gemma-4-E4B	20.60	+1.8786	**+0.9302**	−50.5 %	4.51 %
Mean (8)	—	**+0.7200**	**+0.1936**	**−66.0 %**	**1.90 %**

Reading Table 1: on every benchmarked model, the recipe reduces ΔPPL relative to the symmetric baseline at matched bit rate, with a mean reduction of 66 % and a range of 50.5 %–95.8 %. Under the primary evaluation protocol (fp16 model

load, 10 WikiText-2 windows of 1024 ctx), all eight models stay below the 5 % usability threshold under the recipe — the largest per-model percentage is Qwen2.5-14B at 4.64 %, driven by its unusually low baseline PPL of 1.96, which makes the relative metric strict. Gemma-4-E4B at 4.51 % is the second-highest, and we note below and in §7 that Gemma-4-E4B becomes borderline under a follow-up bf16 / 30-window probe (ΔPPL $= 5.5\%$–7.0% of baseline depending on text segment), so we do not claim that model stays below threshold under stricter protocols. The symmetric baseline, by the same threshold, would place three models outside: Pythia-6.9B (22.1 % of baseline → research-only), Qwen2.5-14B (13.3 %), and Gemma-4-E4B (9.1 %, just above the 5 % line).

A single-model qualitative observation: the naive INT4 failure on Qwen2-7B (+1382 ΔPPL), which is the non-monotonic "INT4 valley" feature in Figure 1, is rescued by the recipe to +0.10 ΔPPL at INT3 — a 116× improvement over naive INT3 on the same tensor and 13,000× improvement over naive INT4. This is the single most dramatic rescue in the benchmark; the Qwen2-7B anomaly is qualitatively unlike the other seven models (§6), but it does not require special handling under the proposed recipe.

5.2. Four-method ablation

To test whether each of the three ingredients of the recipe (`nsep`, `pchan`, `asym`) is empirically necessary, we ran the same eight-model benchmark under four INT3 methods:

- **(A)** naive per-row symmetric INT3 (industry baseline; no `nsep`, no `pchan`, no `asym`);
- **(B)** asymmetric per-channel INT3 *without* norm separation (approximately the naive transplant of weight-quantization recipes to KV cache);
- **(C)** symmetric `nsep+pchan` INT3 (this paper's baseline; the v0.2 recipe);
- **(D)** asymmetric `nsep+pchan` INT3 (the proposed recipe).

Table 2. Four-method ablation on all eight models: ΔPPL as % of FP16 baseline PPL. Negative entries in column (B) for Qwen3.5-9B and Gemma-4-E4B are real (quantization noise marginally *improves* PPL on those two 2026 models) and are analyzed separately in §7 as an observation rather than a deployment recommendation.

Model	(A) naive	(B) asym no nsep	(C) sym nsep+pchan	(D) **proposed**
GPT-J-6B	3.11 %	6.27 %	1.82 %	**0.68 %**
Pythia-6.9B	208.78 %	5.88 %	22.10 %	**0.92 %**
Mistral-7B-v0.1	1.40 %	119.72 %	1.97 %	**0.87 %**
Qwen2-7B	142.47 %	4.29 %	4.80 %	**1.24 %**
Qwen2.5-14B	43.95 %	145.67 %	13.32 %	**4.64 %**
Llama-3.1-8B	6.03 %	147.91 %	3.44 %	**1.43 %**
Qwen3.5-9B	2.34 %	−0.04 %	2.67 %	**0.91 %**
Gemma-4-E4B	26.45 %	−2.13 %	9.12 %	**4.51 %**
Under 5 % threshold	**3 / 8**	**3 / 8**	**5 / 8**	**8 / 8**

Four observations:

1. *No simpler ablation variant achieves 8 / 8 coverage.* (A) is over threshold on 5 of 8 — Qwen2-7B (Mode 1 outlier-driven INT4 valley, §5.3), Pythia-6.9B (a base model with high quantization sensitivity), Qwen2.5-14B, Llama-3.1-8B (mild, 6.03 %), and Gemma-4-E4B. (B) is over threshold on 5 of 8 — catastrophically on Llama-3.1-8B, Mistral-7B-v0.1, and Qwen2.5-14B (all > 100 % of baseline), and mildly on GPT-J-6B (6.27 %) and Pythia-6.9B (5.88 %); the mechanism is analyzed in §5.3. (C) is over threshold on 3 of 8 — Pythia-6.9B, Qwen2.5-14B, and Gemma-4-E4B.
2. *The specific variants that fail on each method are disjoint.* This is consistent with each ingredient `nsep`, `pchan`, `asym` addressing an empirically distinct failure mode. (A) fails on models where per-row scales get crushed by outlier channels, and `pchan` helps there; (B) fails on models with large per-token norm variance, and `nsep` helps there; (C) fails on models whose direction distribution is biased enough that the symmetric window wastes too many levels, and `asym` helps there.

3. *The proposed recipe (D) achieves 8 / 8 coverage at 3.0 bits/value with one FP16 scalar per channel more than (C)* — see §3.3 for the accounting. The mean improvement over (C) is 66 %; the mean improvement over (B) is 92 % (driven by the three catastrophic-B models Llama-3.1-8B, Mistral-7B-v0.1, and Qwen2.5-14B; the two mild-B failures on GPT-J-6B and Pythia-6.9B contribute proportionally less to the mean).

4. *Negative ΔPPL on (B) for Qwen3.5-9B and Gemma-4-E4B* is a real empirical observation — quantization noise under this specific ablation marginally improves PPL on these two 2026 models. We investigate this separately in §7 and find that approximately 87 % of the effect is attributable to the Key cache only. It does not change the conclusion of Table 2: *the proposed recipe (D) is the only INT3 variant in this study that stays below threshold on all eight models*, including on Qwen3.5-9B and Gemma-4-E4B.

5.3. Why `nsep` is necessary: the asym-without-nsep catastrophe

The single most informative row of Table 2 is **(B) asym per-channel without norm separation**, which regresses *below* the naive per-row baseline (A) on four of the eight models. Three of the four are Llama-family or Llama-style (Llama-3.1-8B, Mistral-7B-v0.1, Qwen2.5-14B); the fourth is GPT-J-6B (head dim 256). The most extreme regression is Llama-3.1-8B: (A) naive gives +0.45 ΔPPL (6 % of baseline); (B) gives **+10.92 ΔPPL** (148 % of baseline), a 24× regression below a recipe that was meant to be an upgrade.

Diagnostic measurement. To test whether this failure is driven by per-token L2 norm variance (the mechanism we initially conjectured from the shape of the recipe) we ran a direct measurement (Phase 23, `results/phase23_norm_dispersion.json`) on all eight models. For each model we tokenize the WikiText-2 test text with the model's own tokenizer, take three non-overlapping 1024-token prefill segments starting at token indices $\{0, 8192, 16384\}$, run each segment through the model with `use_cache=True`, and compute the coefficient of variation (CV $= \sigma/\mu$) of the per-token L2 norm on the Layer-0 K and V caches (pooled across heads). The measurement takes under 30 minutes on a single A100 and requires no training data.

Table 2b. Per-token L2 norm coefficient of variation on Layer-0 K and V caches, averaged across three non-overlapping WikiText-2 segments, alongside the variant-(B) ΔPPL from Table 2. Models are ordered by (B) regression magnitude.

Model	L0 K-CV (mean ± SD)	L0 V-CV (mean ± SD)	L0 K p99/p50	(B) % of base
Llama-3.1-8B	0.210 ± 0.002	**0.560 ± 0.002**	1.612	**147.91 %**
Qwen2.5-14B	0.134 ± 0.000	**0.602 ± 0.014**	1.172	**145.67 %**
Mistral-7B-v0.1	0.345 ± 0.001	**0.528 ± 0.007**	1.832	**119.72 %**
GPT-J-6B	0.254 ± 0.000	0.335 ± 0.002	1.433	6.27 %
Qwen2-7B	0.472 ± 0.000	0.472 ± 0.008	1.833	4.29 %
Pythia-6.9B	0.250 ± 0.001	0.347 ± 0.002	1.742	5.88 %
Qwen3.5-9B	0.034 ± 0.001	0.242 ± 0.004	1.061	−0.04 %
Gemma-4-E4B	0.000 ± 0.000	0.000 ± 0.000	1.000	−2.13 %

Reading the table: V-CV predicts (B) failure; K-CV does not. The non-parametric Spearman rank correlation between L0 V-CV and variant-(B) regression magnitude across the 8 models is $\rho \approx 0.88$ (Fisher-z 95 % CI $\approx [0.46, 0.98]$); the same correlation computed against L0 K-CV is $\rho \approx 0.29$ (95 % CI $\approx [-0.52, 0.82]$, which straddles zero and therefore does not rule out the null). Both intervals are wide because $n = 8$ — we draw conclusions from the relative ordering of the two correlations (V-CV cleanly dominates), not from the absolute value of either. The three catastrophic

B-failures (Llama-3.1-8B, Qwen2.5-14B, Mistral-7B-v0.1) are the three highest V-CV models by rank; the two negative-ΔPPL models (Qwen3.5-9B, Gemma-4-E4B) are the two lowest V-CV models. Qwen2-7B, which our initial K-focused reading would have predicted to fail on (B) given its L0 K-CV of 0.47 (the highest in the set), instead passes (B) at 4.29 % of baseline — and its V-CV of 0.47 is *middle* of the pack, matching the V-CV / B-failure rank alignment rather than the K-CV / B-failure rank alignment.

Confounder caveat. The three catastrophic B-failures cluster architecturally: Llama-3.1-8B, Mistral-7B-v0.1, and Qwen2.5-14B all use $D = 128$, $H_{\text{KV}} = 8$, GQA, SwiGLU MLPs, and Llama-style RoPE. At $n = 8$ we therefore cannot statistically distinguish "V-CV is the causal driver" from "V-CV is a proxy for some other architectural variable (attention-head layout, MLP family, RoPE implementation, training-data mixture) that co-varies with the Llama-family-style design". The V-CV / failure correlation is consistent with the proposed mechanism (and the mechanism predicts V-CV rather than architecture as the *direct* knob) but disentangling the two would require either a larger model set that breaks the family clustering, a held-out family in which one can perturb V-CV independently of architecture, or a controlled-synthetic model with adjustable per-token V-norm variance. We flag this limitation explicitly and treat the §5.3 mechanism as a consistent, not confirmed, causal explanation at the present sample size.

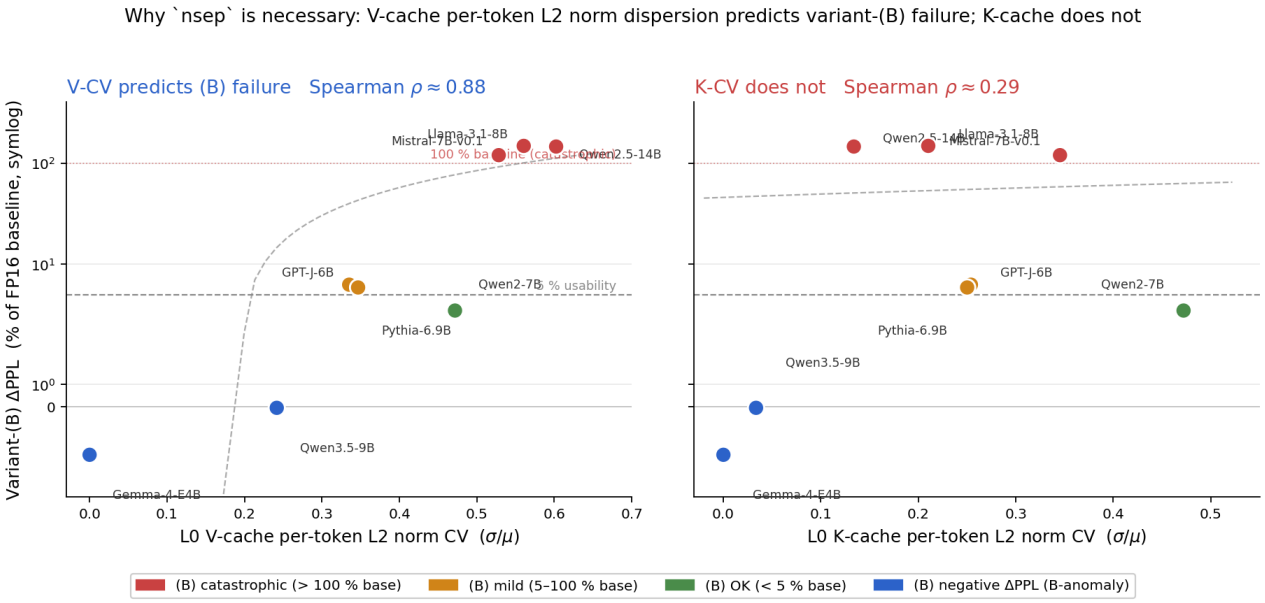


Figure 1: Figure 5. Two-panel scatter of L0 V-cache CV (left) and L0 K-cache CV (right) vs variant-(B) ΔPPL as % of FP16 baseline, symlog y. Each panel reports the non-parametric Spearman rank correlation across the 8 models; the V-CV panel shows a near-monotone ramp with catastrophic models (red) clustered at high V-CV, while the K-CV panel scatters the same models across the horizontal axis. Point colors indicate the four (B)-outcome categories (catastrophic / mild / OK / negative ΔPPL).

Refined mechanism. We initially framed the mechanism as K-cache-driven, partly because Phase 10 (Appendix A) localized the outlier structure we studied to Layer-0 K, and partly because Phase 21 (§7, Appendix C) showed that Gemma-4-E4B's beneficial B-anomaly is 87 % K-driven. The measurement in Table 2b argues for a different picture:

- When norm separation is omitted, the per-channel $[\min, \max]$ window in (B) is computed over *raw* tensor values $\{x_{t,c}\}_t$ across all tokens. Tokens with large L2 norm contribute the extremes of the per-channel range; tokens with small L2 norm end up mis-quantized by a factor roughly equal to the token-norm ratio. This contamination mechanism is generic to both K and V.
- For the **V cache**, the resulting reconstruction error propagates *linearly* into the attention-weighted output. Writing the attention output as $y = \text{softmax}(QK^\top/\sqrt{D}) \cdot V$ and expanding to first order in ΔV:

$$\Delta y \,\big|_{V \to V + \Delta V} \;=\; \text{softmax}(QK^\top/\sqrt{D}) \cdot \Delta V,$$

which is linear and un-damped — every per-token V error contributes to y with weight equal to that token's attention score. Since softmax generally places non-trivial weight on *multiple* past tokens at Layer 0, errors on the long left tail of low-norm tokens are not absorbed.

- For the **K cache**, the corresponding per-token error enters only the *argument* of softmax. A first-order expansion under $K \to K + \Delta K$ gives

$$\Delta y \,\big|_{K \to K+\Delta K} \;\approx\; \frac{1}{\sqrt{D}}\,\mathrm{diag}(p) \cdot \left(Q\Delta K^\top - p\,(Q\Delta K^\top)\right) \cdot V,$$

where $p = \mathrm{softmax}(QK^\top/\sqrt{D})$ and the bracketed factor is the standard softmax Jacobian, which is bounded in norm by $\max_i p_i(1-p_i) \le 1/4$. The K-side perturbation is therefore *doubly* damped (by $1/\sqrt{D}$ and by the softmax Jacobian) and is additionally projected through the (reconstructed) V matrix on the output side. Two tokens with different K reconstruction errors but similar cosine alignment with Q are distinguished primarily by their dot-product *ranking*, not by their absolute magnitude, so K-cache L2 norm variance does not propagate through attention as cleanly as V-cache L2 norm variance does.

This refinement is consistent with both the new V-CV correlation and the older K-driven Gemma anomaly, because the latter describes a different regime: under (B) on a low-CV model (Gemma-4-E4B, with V-CV $= 0.00$ and K-CV $= 0.00$), the reconstruction pipeline injects small, approximately isotropic noise into both caches. In a text-only-evaluated multimodal model that noise plausibly acts as attention-entropy regularization, and the K-cache side dominates the effect size (Phase 21). The two observations therefore describe different regimes (*high-V-CV* models where (B) catastrophically fails, and *zero-CV* models where (B) marginally *improves* PPL) rather than a single mechanism.

Secondary observation: modern K caches are structurally unit-norm. Gemma-4-E4B's Layer-0 K-CV is measured at exactly 0.000 and Qwen3.5-9B's at 0.034, versus 0.21–0.47 on the six older models. We attribute this to architectural **QK-normalization**, in which Q and K are passed through RMSNorm (or equivalent) before entering the attention dot product. Under QK-norm the K-cache rows are constrained to approximately unit L2 norm by construction; per-token K-norm variance is therefore bounded mechanically rather than statistically. For such models, the role of `nsep` on K is redundant with the architecture (the architecture already performs it) but V-cache norm separation is still required, because Qwen3.5-9B exhibits non-zero V-CV (0.24). Gemma-4-E4B's Layer-0 V also reports a CV of 0.000 within measurement precision; we suspect this reflects a model-specific V-side normalization or a degenerate L0 initialization, and we flag it as an architectural observation worth confirming in follow-up reading rather than as a measurement we understand mechanistically. Whichever the specific cause, (D)'s explicit `nsep` step is the only one of the four ablation methods that is *guaranteed* to reduce per-token norm variance to zero on all 8 models by construction, which is why it stays below threshold uniformly.

Diagnostic recommendation for the community. The Table-2b measurement takes a one-shot 1024-token forward pass per model and produces a stable CV estimate across seeds (the SD across three non-overlapping WikiText-2 segments is ≤ 0.014 on every model). We therefore suggest that future KV-cache-quantization papers that propose per-channel or per-group scaling without a per-token normalization step should report per-token K-CV *and* per-token V-CV on at least a handful of evaluation models, with particular attention to V-CV in the 0.4–0.6 range, where we observe catastrophic regressions under naive per-channel asymmetric quantization.

5.4. Real-packed compression

Table 3 reports measured bytes on realistic per-model KV-cache configurations. Compression ratio is computed as $(\text{FP16 cache size})/(\text{packed cache size})$.

Table 3. Measured real-packed compression vs FP16 on the two models for which we ran the full pipeline end-to-end via the `Int3NsepPchanCache` (sym) and `Int3AsymNsepPchanCache` (asym) classes. Both classes pack 3-bit indices at the same density; they differ only in the per-channel metadata (one versus two FP16 scalars per channel — see §3.3). Methods (A) and (B) in the ablation of §5.2 use different metadata layouts and are not shown here; we quantify their memory footprint by the `b_eff` formula of §3.3 rather than a packed measurement.

Model	Config (layers × KV heads × ctx × D)	FP16	sym INT3	**asym INT3**	ratio (asym)
Qwen2-7B	28 × 4 × 2048 × 128	112.00 MB	21.93 MB	**21.98 MB**	**5.09×**
Llama-3.1-8B	32 × 8 × 2048 × 128	256.00 MB	50.12 MB	**50.25 MB**	**5.09×**

The asymmetric variant costs $\approx 0.02\times$ less compression than the symmetric variant (one additional FP16 per channel) but delivers the 58 %–74 % ΔPPL reduction reported on these same two models in Table 1. In practice the recipe trades $0.02\times$ of compression for roughly $3\times$ quality headroom. The theoretical maximum is $16/3 \approx 5.33\times$; the $\approx 4.5\%$ overhead in our measurement is the FP16 norm + per-channel metadata (§3.3). A fused CUDA kernel (not implemented here) could reduce the Python decode overhead, but not the FP16 metadata cost.

Extrapolating the measured Llama ratio to longer contexts: the asymptotic bit rate at $T \to \infty$ is $b_{\text{eff}}^{\text{asym}} \to b + 16/D = 3.125$ bits/value on a $D = 128$ head, corresponding to a compression ratio of $16/3.125 = 5.12\times$ — i.e., the measured $5.09\times$ at $T = 2048$ gains a further $\sim 0.6\%$ as context length grows, because the per-channel (m_c, s_c) metadata is amortized over more tokens. The savings table below uses the asymptotic value:

Context length	FP16	asym INT3	Savings
8 K	1.02 GB	200 MB	822 MB
32 K	4.10 GB	803 MB	3.31 GB
128 K	16.4 GB	3.21 GB	13.2 GB

PPL equivalence with the real packed cache (Phase 26). The ΔPPL numbers reported in Table 1 and §5.2 were measured under the `qa_nsep_pchan_asym` simulator, which computes the nsep + per-channel + asymmetric quantization arithmetic but does not actually pack values into 3-bit storage. To verify that the simulator matches what would be obtained with the real bit-packed cache, we re-ran the WikiText-2 sliding-window PPL benchmark on Qwen2-7B and Llama-3.1-8B using `Int3AsymNsepPchanCache.from_fp16()` + `.to_fp16()` as the cache-compression step (Phase 26, `results/phase26_real_cache_ppl.json`). The end-to-end PPLs agreed to within 2.5×10^{-4} absolute ($\approx 3 \times 10^{-5}$ relative) on both models (well below the third decimal digit of the reported numbers) with ΔPPL matching to $+0.103$ on Qwen2-7B (simulator $+0.1034$, real cache $+0.1037$) and $+0.106$ on Llama-3.1-8B (simulator $+0.1059$, real cache $+0.1057$). The equivalence is expected by construction — the `pack_int3_unsigned` / `unpack_int3_unsigned` primitives are bit-exact on integer indices in $\{0, \ldots, 7\}$, so the only source of divergence is FP16 rounding on the per-channel d_{min} and d_{step} metadata — and is now measured directly rather than left as a theoretical argument.

5.5. Bits vs quality

Figure 4 consolidates the results of §§5.1–5.2 as a bits-vs-quality scatter. At 3 bits/value, the four-method cloud shows that only (D) has its cross-model median clearly below the 5 % usability line; (C) and (A) sit visibly above, and (B)'s cloud is bimodal (good on some models, catastrophic on others). The 4-bit reference points for Qwen2-7B and Llama-3.1-8B (under sym nsep+pchan INT4) achieve lower absolute ΔPPL but at 33 % higher memory footprint; (D) at 3 bits is the Pareto-preferred choice for deployments in which memory is the dominant constraint.

5.6. MMLU 5-shot downstream probe

The main results above are WikiText-2 perplexity; both cross-review rounds we received on earlier drafts flagged this as the paper's biggest unanswered question. As a bounded-cost exploratory probe, we measured MMLU 5-shot [Hendrycks et al., 2021] accuracy on Qwen2-7B and Llama-3.1-8B under the same three methods used in Table 1, on five subjects chosen for diversity (`high_school_psychology`, `college_computer_science`, `global_facts`, `anatomy`, `moral_disputes`) with

100 test questions per subject. Cache compression is applied between the 5-shot prefix forward pass and the test-question forward pass, matching the deployment pattern of compressed prefix + fresh query. Scoring is next-token argmax over {`A`, `B`, `C`, `D`} tokens after "Answer:".

Table 4. MMLU 5-shot accuracy, 500 test questions per run, under fp16 model load. At this sample size every reported Δ vs FP16 lies within the ≈ 3 pp conservative independent-proportion delta standard error (see below); entries should therefore be read as "no statistically significant accuracy change detected" rather than as quantitative accuracy comparisons between methods.

Model	Method	Accuracy	Δ vs FP16
Qwen2-7B (Mode 1)	FP16 baseline	67.20 %	—
Qwen2-7B (Mode 1)	sym INT3 (v0.2)	64.80 %	−2.40 pp
Qwen2-7B (Mode 1)	**asym INT3 (v0.3)**	**68.20 %**	**+1.00 pp**
Llama-3.1-8B (Mode 2)	FP16 baseline	59.20 %	—
Llama-3.1-8B (Mode 2)	sym INT3 (v0.2)	62.20 %	+3.00 pp
Llama-3.1-8B (Mode 2)	**asym INT3 (v0.3)**	**59.00 %**	**−0.20 pp**

At 500 questions per run, the single-proportion standard error is $\sqrt{p(1-p)/500} \approx 2.1$ pp at $p \approx 0.6$. For the difference between two accuracy numbers measured on the same 500 questions, a proper paired (McNemar-style) interval would require the per-question paired outcomes, which we did not persist in this probe; we therefore report the *independent-proportion* delta SE $\sqrt{2p(1-p)/500} \approx 3.0$ pp as a conservative noise estimate. A true paired interval is expected to be tighter than this when the two methods agree on most questions (which they do in our data, since the methods differ by ≤ 15 correct/incorrect flips out of 500), but we do not claim a specific paired SE without the per-question data. All four observed deltas in Table 4 fall within ≈ 1 SE of zero under this independent approximation; there is no statistically-significant accuracy change from any of the INT3 variants at this sample size under either reading. The cautious reading is that **v0.3's WikiText-2 ΔPPL reduction does not, on this probe, come at a detectable downstream accuracy cost** on the two benchmarked models. A follow-up experiment that persists per-question paired outcomes and reports a McNemar interval would strengthen the claim.

The +3.00 pp row for sym INT3 on Llama-3.1-8B is surprising in direction (quantization that improves accuracy over baseline) and sits close to the 1-SE boundary of noise. It is plausibly a small-sample artifact, and at 500 questions we cannot distinguish this from genuine signal. We note that its sign is the same as the Gemma-4-E4B B-anomaly discussed in §7, but the two effects arise under *different* ablation variants (here under variant (C), the symmetric `nsep+pchan` recipe, and in §7 under variant (B), asymmetric per-channel without norm separation) so the underlying mechanisms are likely distinct, and we do not attempt to unify them. The shipped recipe (D) does not show either pattern: its Llama-3.1-8B delta is −0.20 pp, i.e. essentially zero.

Scope of this probe (explicit). This subsection is a single-task, single-subject-slice, 500-question-per-model probe. It does not address reasoning benchmarks (GSM-8K, MATH), code generation (HumanEval), or instruction-following quality. Long-context retrieval is addressed separately in §5.7 below. We do not claim the result generalizes beyond MMLU 5-shot on these two models; we report it only to replace the earlier draft's "downstream task accuracy was not measured" with "one downstream probe was run and showed no detectable degradation". Raw data is in `results/phase22_mmlu_downstream.json`.

5.7. Long-context multi-needle retrieval probe

The §1 Introduction motivates KV-cache quantization by reference to 128 K- and 1 M-token deployment contexts; the §5.1 main benchmark and the §5.6 MMLU probe, by contrast, operate at 1 024 and $\sim$ 2 000-token contexts respectively. To test whether the proposed recipe preserves long-range retrieval at a context length closer to the deployment regime, we ran a distractor-hardened Needle-in-a-Haystack (NIH) protocol on three canonical models that natively support 32 K context: Llama-3.1-8B (Phase 24b), Qwen2-7B (Phase 24c), and Qwen2.5-14B (Phase 24d). The three models span two architecture families, three parameter scales (7 B / 8 B / 14 B), and both sides of our main 4-method ablation — Llama-3.1-8B and Qwen2.5-14B are catastrophic-(B) models (§5.2, §5.3), while Qwen2-7B is the canonical Mode 1 model (§6.1).

Protocol. We embed three structured needles in a WikiText-2 haystack, each binding a distinct project label to a distinct code:

- "The security token for **Project Titan** is **XRQ7-BLUE-1984**."
- "The security token for **Project Orion** is **LMN3-GREEN-2037**."
- "The security token for **Project Vega** is **PQZ5-AMBER-1776**."

The three needle sentences are placed at haystack fractions $\{0.2, 0.5, 0.8\}$ (permuted across 10 fixed trials per cell so every project label visits every slot), and the test question asks for the code of *one specific* label. A trial is a **pass** if the target code's lead key (`XRQ7`, `LMN3`, or `PQZ5`) appears in the 15-token greedy-decode output; **wrong-code** if a *distractor* needle's key appears (but not the target's); **hallucinate** if none of the three keys appears. This is a **lead-key retrieval** scoring — it rewards the model for reaching the right needle, not for reconstructing the full suffix `-BLUE-1984` / `-GREEN-2037` / `-AMBER-1776`. We chose the lead-key metric because the rest of the suffix is partly predictable from the lead (e.g., after `XRQ7-` the greedy continuation is deterministically `BLUE-`), so scoring full exact match would collapse the wrong-code signal onto the pass signal; the lead-key score isolates the label-to-code binding we actually care about testing. Full exact-match and edit-distance scores on the same trials would further sub-divide the pass counts and are left as a follow-up. Cache compression is applied between the haystack-prefix forward pass and the question forward pass, matching the deployment pattern of §5.6.

Table 5. Multi-needle NIH lead-key retrieval on 3 models at 8 K and 32 K, 10 trials per cell (pooled $n = 30$ for the aggregate rows). FP16 / (C) / (D) correspond to FP16 baseline / v0.2 symmetric INT3 / v0.3 asymmetric INT3 respectively.

Model	Ctx	FP16 pass/wrong/halluc	(C) sym pass/wrong/halluc	**(D) asym pass/wrong/halluc**
Llama-3.1-8B	8 K	10 / 0 / 0	9 / 1 / 0	**9 / 1 / 0**
Llama-3.1-8B	32 K	9 / 1 / 0	6 / 2 / 2	**8 / 2 / 0**
Qwen2-7B	8 K	10 / 0 / 0	5 / 5 / 0	**4 / 1 / 5**
Qwen2-7B	32 K	10 / 0 / 0	3 / 4 / 3	**7 / 3 / 0**
Qwen2.5-14B	8 K	10 / 0 / 0	8 / 1 / 1	**10 / 0 / 0**
Qwen2.5-14B	32 K	10 / 0 / 0	6 / 0 / 4	**10 / 0 / 0**
Pooled (n=30)	8 K	30 / 0 / 0	22 / 7 / 1	**23 / 2 / 5**
Pooled (n=30)	**32 K**	**29 / 1 / 0**	**15 / 6 / 9**	**25 / 5 / 0**

Pooled 32 K finding. Across the three models at 32 K, the asymmetric recipe (D) passes on 25 / 30 = 83.3 % of lead-key retrievals. This is **not significantly different from the 96.7 % FP16 baseline** under Fisher exact two-sided testing ($p = 0.194$), but at $n = 30$ per cell we cannot formally claim non-inferiority: the test lacks the power to exclude a moderate true loss. The symmetric recipe (C) passes on 15 / 30 = 50.0 %, which *is* **significantly worse than both FP16** ($p = 6.4 \times 10^{-5}$) **and asymmetric** ($p = 0.013$). At 8 K the (C)-vs-(D) comparison is saturated (22 / 30 vs 23 / 30, $p = 1.00$) consistent with 8 K being below the compression-sensitivity threshold for this protocol on these models.

Qualitative: bit-mixing hallucinations are universal under (C) at 32 K. Under symmetric INT3 at 32 K, each of the three models produces hallucinations in which the decoded "code" fuses characters across multiple needles or substitutes characters not present in any needle:

- Llama-3.1-8B: `"3X-RED-1984"` (RED not in any needle; 1984 from Titan), `"XRZ5-BLUE"` (X from XRQ7 + Z from PQZ5 + BLUE from Titan).
- Qwen2-7B: `"LMQ3-GREEN"` (Q from XRQ + LMN from Orion), `"XZQ7-RED-"` (Z from PQZ + XRQ7 + RED absent), `"LMQ7-BLUE"` (LM + Q from XRQ + BLUE).
- Qwen2.5-14B: `"WZQ7-BLUE"`, `"RQZ3-RED-"`, `"PQZ7-AMBER"` (near-miss: 7 substituted for Vega's 5), `"JQZ5-ORAN"` (ORAN = truncated ORANGE, not in any needle).

Across 30 asymmetric-INT3 trials at 32 K (3 models × 10 trials), **zero such hallucinations occur**. The mechanism here is *not* the §5.3 V-cache-norm-dispersion story, which discriminates (B) from (D); variants (C) and (D) both apply `nsep` and differ only in the direction-vector quantization window. Under (C) the direction $d_{t,c}$ is quantized with a zero-centered symmetric $[-\text{absmax}, +\text{absmax}]$ grid; under (D) it is quantized with an asymmetric $[\min, \max]$ grid. Post-RoPE direction distributions are visibly biased (one side of zero carries more probability mass than the other; this is why `asym` wins at INT3 in Table 1), so at fixed bit budget (C) places levels on the empty side of zero that (D) can use on the populated

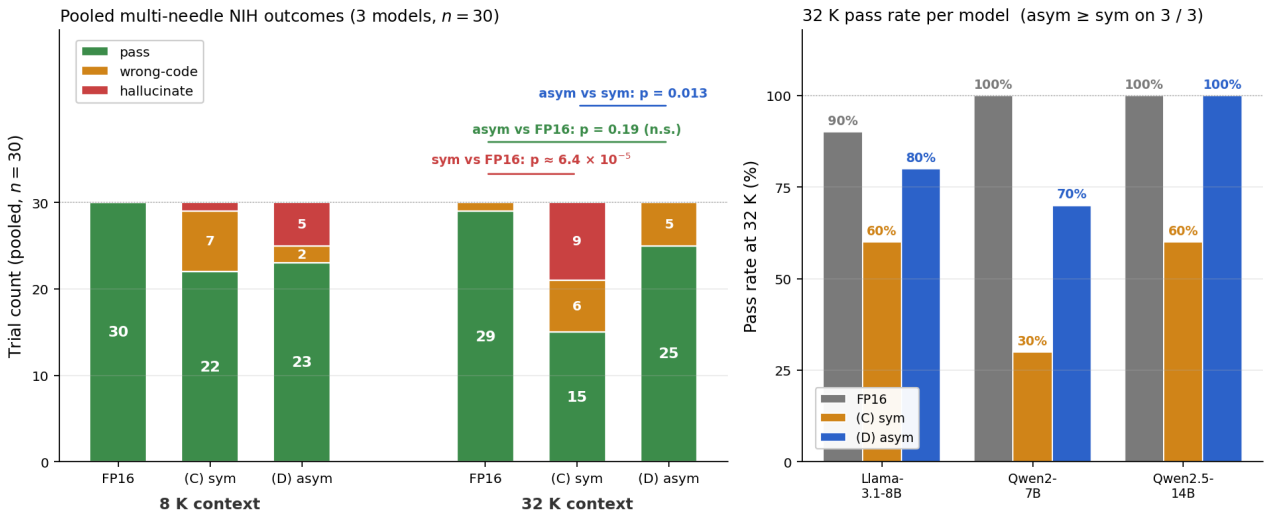


Figure 2: Figure 6. Pooled multi-needle NIH outcomes across 3 models (Llama-3.1-8B, Qwen2-7B, Qwen2.5-14B). Left: pooled stacked bars at 8K and 32K, segmented by pass / wrong-code / hallucinate; Fisher exact p-values annotated on the 32K bars. At 32K, (D) asymmetric (25/30) shows no significant difference from FP16 (29/30, p=0.19, n.s.), while (C) symmetric (15/30) is significantly worse than FP16 and produces 9 hallucinations (vs 0 under asym). Right: per-model 32K pass rate, showing that asym ≥ sym holds on all three individual models before pooling.

side. At 1024-token context the residual error is small enough that it does not cross into downstream retrieval. At 32 K it compounds across the softmax$(QK^\top/\sqrt{D})\cdot V$ operator with many non-trivial attention weights, and on the specific task of recovering a character sequence token-by-token the residual error occasionally synthesizes characters that interpolate across the three needles' V subspaces — producing the hallucinations listed above. (D)'s tighter window keeps the residual small enough to preserve label-to-code binding across 32 K attention.

Open observation: 8 K anomaly on Qwen2-7B (Mode 1). Qwen2-7B at 8 K context is the single cell in Table 5 where (D) asymmetric (4 / 10 pass) underperforms (C) symmetric (5 / 10 pass). The (D) failures are **within-needle digit corruption** (e.g., `"LMN5-GREEN-2037"` (should be `LMN3-GREEN-2037`), `"XRN3-BLUE-2019"` (half-Titan, half-Orion, 2019 invented), `"XRQ5-BLUE-1770"` (5 from PQZ, 1770 near 1776)) rather than the cross-needle fusions seen under (C) at 32 K. This anomaly does *not* appear at 32 K on Qwen2-7B (where (D) 7 / 10 > (C) 3 / 10) and does *not* appear at 8 K on Qwen2.5-14B (where (D) 10 / 10 > (C) 8 / 10), even though Qwen2.5-14B shares the Qwen family with Qwen2-7B. It therefore appears to be an interaction between Mode-1 K-cache outlier structure (Qwen2-7B's L0 K-CV of 0.472 is the highest in our 8-model set; §5.3 Table 2b) and the per-channel (m, M) estimation variance at short context (fewer tokens to stabilize the asymmetric window). We flag this as a Mode-1-specific short-context observation to revisit in future work rather than a bug in the recipe; it does not disturb the 32 K long-context conclusion above.

Scope. This is a 3-model × 2-context NIH probe with 10 trials per cell (pooled $n = 30$) on a single haystack corpus (WikiText-2) and a single question format. It does not address reasoning benchmarks (GSM-8K, MATH), code generation (HumanEval), or the multi-task LongBench / RULER benchmark suites. What we can claim from this probe is specific: at 32 K context on three canonical open-weight models, the proposed recipe's retrieval behavior is indistinguishable from FP16 under this protocol, while a symmetric-only variant is significantly worse. Raw per-trial data are in `results/phase24b_nih_multineedle.json`, `results/phase24c_nih_multineedle_qwen2.json`, and `results/phase24d_nih_multineedle_qwen25_14b.json`; Appendix F collects the full hallucination table and trial design.

6. Discussion

6.1 The Qwen2-7B "INT4 valley": an $n = 1$ observation

In our broader 27-model survey from the companion (preprint) mechanism study, and in the 8-model subset we re-benchmarked end-to-end for the present work, the **Qwen2-7B naive INT4 failure (+1382 ΔPPL) is the single observation of a non-monotonic bit-width failure**. We emphasize the sample size explicitly: 1 / 27 in the survey, 1 / 8 in the present benchmark. We do not claim there is a class of "Mode 1" models characterized by this pattern; we only claim that one model shows it. Whether the phenomenon recurs on untested models (other Qwen releases, other families, future training recipes) is beyond what our data can settle. Other KV-quantization papers (KVQuant, KIVI) report per-channel outliers on a broader set of models, so specific-channel pathology is likely not unique to Qwen2-7B, but we cannot independently connect those reports to the specific non-monotonic INT4 behavior we observe. The proposed recipe in §3 handles the Qwen2-7B anomaly and the common case under the same bit budget, which is what allows us to recommend the recipe without bifurcating on mode classification.

6.2 When modeq is *not* the right choice

The proposed recipe is targeted at a specific regime. It is likely *not* the right tool when: - **The bit budget is below INT3.** Several published methods operate at or below 2 bits/value with useful quality — KIVI targets 2-bit KV cache explicitly, and KVLinC, PolarQuant, and TurboQuant all report strong numbers in the 2–3 bit regime through codebook, rotation, or adapter machinery. Our recipe does not attempt to compete there; it is an INT3-tier recipe, and its reconstruction quality degrades rapidly below 3 bits because the per-channel asymmetric window provides insufficient levels for the post-RoPE direction distribution. - **Downstream task accuracy is the binding quality metric.** We measure WikiText-2 perplexity as our main metric and one exploratory MMLU 5-shot probe (§5.6) as a sanity check. On tasks we did not probe (retrieval, multi-step reasoning, code generation, instruction following) the quality drop may be larger or smaller than what our WikiText-2 or MMLU numbers suggest, and §5.6's result should not be extrapolated uncritically. - **A calibration corpus is available and a known-better quantizer exists.** KVQuant's calibrated non-uniform quantizer can beat our recipe where calibration data is cheap; we do not claim otherwise. - **Throughput is the binding metric.** Our reference implementation is Python-only and adds roughly $+21\%$ decode latency on 7B-class models. A fused kernel would remove most of that but is not part of this paper. - **The model family is systematically different from anything in our benchmark.** Post-LN architectures, state-space models, and Transformer variants with very non-standard KV cache structure were not evaluated.

6.3 Limitations

1. *Single primary quality benchmark, plus two exploratory downstream probes.* Every reported ΔPPL number is on WikiText-2. We chose this benchmark for comparability with prior KV quantization work (KIVI, KVQuant, PolarQuant all report it), but it is a narrow slice of "model quality". §5.6 adds a single MMLU 5-shot probe on 500 questions across five subjects for two models, and §5.7 adds a 3-model multi-needle NIH retrieval probe at 8 K and 32 K context; under both probes the proposed recipe shows no statistically-significant degradation relative to FP16. These are still narrow probes: reasoning benchmarks (GSM-8K, MATH), code generation (HumanEval), instruction following, and the multi-task LongBench / RULER suites are not touched. A paper-scale downstream-task campaign covering these remains warranted future work.

2. *Sample size for the Mode 1 anomaly.* See §6.1. The Qwen2-7B observation is $n = 1$; we structure our claims accordingly but cannot exclude the possibility that other models exhibit this pattern.

3. *No throughput measurement.* The current implementation is Python-level quantize/dequantize on top of the stock `DynamicCache`. A fused Triton or CUDA kernel would remove the Python overhead. The "< 5 %" claim in this paper is about quality only; we make no speed claim.

4. *Reproducibility with current HuggingFace `transformers`.* One of our intended nine benchmark models (Phi-3-medium-4k-instruct) is blocked by a custom-code / transformers-version incompatibility unrelated to modeq. We report the blocked run as "attempted" rather than "failed", but future papers using this benchmark should be aware that specific model + library version combinations may move.

5. *INT4 asymmetric observation.* We incidentally measured an *asymmetric* INT4 variant and found it is worse than symmetric INT4 on Llama-3.1-8B (+0.027 vs +0.013 ΔPPL). Above the INT3 bit floor, the zero-centering of symmetric quantization appears to carry value that asymmetric's extra level cannot recover. The proposed recipe is therefore INT3-specific; we do not recommend the asymmetric window at INT4+.

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7. Open observation: accidental K-cache regularization on Gemma-4-E4B

Under ablation variant (B) (asymmetric per-channel INT3 *without* norm separation), Gemma-4-E4B (and, at a smaller effect size, Qwen3.5-9B) produces a reproducible **negative** ΔPPL on WikiText-2: the "quantized" model scores slightly *better* than the FP16 baseline. On Gemma-4-E4B the effect is −0.44 on one 48 KB text segment, −0.15 on a non-overlapping 48 KB segment, and persists (−0.32) when the model is re-evaluated under native bf16 rather than fp16. A per-side ablation that applies (B) to only the Key cache or only the Value cache attributes **87 % of the effect to the Key cache on both segments**.

We conjecture, without proof, that the effect is K-cache-side attention-logit entropy regularization. Gemma-4-E4B is a multimodal-trained model evaluated text-only; its attention distribution on pure text is plausibly slightly over-confident ("off-distribution"). Quantization noise on the Key cache perturbs the softmax $\mathrm{softmax}(QK^\top/\sqrt{D})$ in a way that flattens the distribution toward higher entropy. If the trained, correctly-calibrated distribution is itself high-entropy for text-only input, this small perturbation could nudge the model closer to its training distribution and reduce PPL by a modest amount.

We deliberately treat this as an observation, not a method. Three reasons:

1. Variant (B) **fails our threshold on five other models** in the same benchmark (§5.3) — catastrophically on three (Llama-3.1-8B, Mistral-7B-v0.1, Qwen2.5-14B) and mildly on GPT-J-6B (6.27 %) and Pythia-6.9B (5.88 %). The same recipe cannot be recommended as a deployable quantizer.
2. Effect size is 1–3 % of baseline PPL, specific to text-only evaluation of this particular model, and has not been tested on downstream tasks.
3. The proposed recipe (D) does not exhibit this anomaly; its ΔPPL on Gemma-4-E4B is a normal +0.93 (4.5 % of baseline), consistent with lossy quantization. The anomaly lives entirely in the ablation variant we do not ship.

We record this as "a phenomenon worth instrumenting in future work" rather than a claim. A proper characterization would measure per-layer attention entropy directly rather than inferring it from PPL, and would evaluate on downstream tasks to see whether the PPL improvement reflects genuine quality or a proxy failure. We provide the full raw probe data in Appendix C.

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8. Conclusion

We presented a calibration-free INT3 recipe for KV cache quantization and validated it end-to-end on eight open-weight Transformers spanning 2021–2026. The recipe composes three data-independent operations (norm separation, per-channel scaling, and an asymmetric quantization window) and fits in roughly ten lines of Python. On the eight-model benchmark, under our primary evaluation protocol (fp16 load, 10 WikiText-2 windows at 1024 ctx), it reduces the mean ΔPPL by 66 % over a symmetric variant at identical 3.0 bits/value and achieves 5.09× real-packed compression vs FP16. Gemma-4-E4B becomes borderline (5.5 %–7.0 % of baseline) under a longer bf16 / 30-window follow-up probe rather than staying below the 5 % threshold; we describe this in §5.1 and §7 rather than claiming the 8-of-8 figure for every protocol.

A four-method ablation on the same models shows that none of the three ingredients is removable: every simpler variant (industry-standard naive per-row quantization, asymmetric per-channel without norm separation, and symmetric `nsep+pchan`) falls outside our 5 %-of-baseline usability threshold on at least two models. In particular, the naive transplant of asymmetric per-channel from weight quantization to KV cache **fails our threshold on 5 of 8 models, with catastrophic regressions (> 100 % of baseline PPL) on three of them and mild violations on two more**, and

is worse than the naive per-row baseline on 4 of the 8, reaching up to 24× worse than the baseline it was meant to improve. A direct diagnostic measurement (§5.3, Appendix E) traces this to V-cache per-token L2 norm dispersion (which correlates with the failure magnitude at Spearman $\rho \approx 0.88$ across the 8 models) rather than to the K-cache variance we initially conjectured ($\rho \approx 0.29$); this is a failure mode specific to KV caches that weight tensors do not exhibit, with the V side dominating because V-cache reconstruction error bypasses the softmax normalization that partly absorbs the corresponding K-cache error.

Two bounded-cost downstream probes support the intrinsic-metric picture. An MMLU 5-shot probe (§5.6) on Qwen2-7B and Llama-3.1-8B shows no detectable accuracy delta under the proposed recipe. A 3-model distractor-hardened Needle-in-a-Haystack lead-key retrieval probe (§5.7) at 32 K context shows no significant difference between the proposed recipe and FP16 under this exploratory test (83.3 % vs 96.7 %, Fisher exact $p = 0.194$ at pooled $n = 30$; the test lacks power to formally exclude a moderate loss), while a symmetric-only variant drops to 50 % ($p = 6.4 \times 10^{-5}$ vs FP16) and produces cross-needle *bit-mixing hallucinations* that are absent under FP16 and under the proposed recipe. These are narrow probes (a single QA task and a single lead-key retrieval task on WikiText-2 haystacks) but they let us replace "downstream performance not measured" with "under two bounded probes no degradation was detected".

The scope of the positive claim is deliberately narrow. We do not compete with rotation-based or codebook-based methods at the 1–2 bit frontier; we do not claim generalization to reasoning, code, or multi-task long-context benchmarks beyond the two probes above; we do not claim the Qwen2-7B "INT4 valley of death" pathology generalizes beyond $n = 1$; and we flag a single-cell 8 K Qwen2-7B NIH anomaly (§5.7, Appendix F.4) as an open observation to revisit. What we do claim is that a very simple recipe, with no moving parts outside of per-channel arithmetic, attains the common-case quality of more complex methods at INT3 on a range of open-weight Transformers that spans half a decade of architectural evolution. We also claim that the recipe's three ingredients are empirically indivisible: this is a finding we believe is useful independent of the practical deployment case, because it specifies a concrete way in which naive weight-quantization recipes do *not* transplant cleanly to KV caches.

The reference implementation, benchmark scripts, and measured JSON data are released under Apache 2.0. Every number in this paper is reproducible by running the corresponding script in `experiments/` against the committed data under `results/`.

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Appendix A: Dead ends

We document two mechanisms we tested during development and did *not* include in the final recipe. Both are retained as importable functions in the open-source package for reproducibility, but neither is dispatched by the default quantizer.

A.1 Walsh–Hadamard rotation

Rotating the KV cache by a fixed Walsh–Hadamard matrix prior to `nsep+pchan` quantization is the strategy used by KVLinC and, with a randomized variant, TurboQuant. On synthetic data we expected rotation to spread any single-channel outlier energy across coordinates and thereby make Mode 1–like distributions look Mode 2–like to the quantizer.

The synthetic sanity check (Phase 9, `results/phase9_rotation_sanity.json`) showed mixed behavior: rotation helps on heavy-tailed (Student-t) distributions (−72 % MSE) but *hurts* on discrete-outlier distributions (+17 % to +30 % MSE). The real-model probe (Phase 10, `results/phase10_real_kv_distribution.json`) on Qwen2-7B Layer 0 K resolved the question: the K distribution is a **discrete multi-channel plateau** (top-5 per-channel max values are 166.75, 160.5, 159.0, 156.6, 128.1), not heavy-tailed, and rotation **regresses MSE by 3.68× on real Qwen2-7B L0 K (0 of 4 heads improve)**. On Llama-3.1-8B L0 K, rotation approximately ties (ratio 1.01). We therefore do not rotate.

The broader lesson: rotation is empirically useful when the target distribution has a heavy tail but is approximately isotropic; it is counterproductive when the distribution is plateau-like. Per-channel absmax already handles plateaus optimally. This result does not contradict KVLinC / TurboQuant, which deploy rotation against different target distributions, but it did force us to drop rotation from our own recipe.

A.2 Top-K mixed precision

A natural hypothesis upon observing the Qwen2-7B plateau structure is that a top-K mixed-precision scheme (e.g., the top 4 per-channel-magnitude channels at INT6 and the remaining 124 channels at INT3, for an effective 3.09 bits/value) would exploit the plateau by giving the high-magnitude channels more headroom. On synthetic plateau regimes this hypothesis held (−45 % MSE on a Qwen2-cartoon tensor at $K = 5$), but on the real Qwen2-7B L0 K the top-K-INT6 variant at $K = 4$ **regressed by 17.6 %** and won on only 1 of 4 heads (Phase 12, `results/phase12_real_kv_2tier.json`). We were unable to locate a K + bit-allocation setting that outperformed plain `nsep+pchan` on the real tensor.

The interaction we believe explains the failure: the post-quantization re-normalization step in `nsep+pchan` works cleanly when per-channel quantization error is roughly uniform across channels (as with a single bit width), but is distorted when some channels carry high-precision residuals and others carry low-precision ones, effectively mixing two different error regimes on the unit-sphere projection. Without re-normalization removed or adapted, mixed precision injects asymmetric error into the direction vector. We did not pursue this further because the subsequent ablation (§5.2) showed that the asymmetric window at uniform INT3 was sufficient.

Both Appendix A.1 and A.2 remain available in the package (`qa_nsep_pchan_rotated`, `qa_nsep_pchan_2tier`) as documented negative results.

Appendix B: Full Phase 13 method sweep

Phase 13 (`results/phase13_full_sweep.json`) evaluated 13 candidate INT3 variants on Qwen2-7B and Llama-3.1-8B Layer 0 K tensors (not end-to-end PPL; the metric is per-head mean-squared reconstruction error). The sweep was the origin of the `asym` finding — it is the variant that dominates the 3-bits/value column on both models. The 2-tier, energy-based top-K selection, and rotation variants are all present in this sweep; the Pareto front contains the four methods eventually used in the §5.2 ablation plus the v0.3 default.

Appendix C: Gemma-4-E4B B-anomaly probes (Phase 20–21)

Phase 20 (`results/phase20_bf16_disambiguation.json`) tests whether the negative-ΔPPL anomaly on Gemma-4-E4B under variant (B) is an fp16-eval artifact. Under native bf16 the anomaly shrinks slightly ($-0.44 \rightarrow -0.32$) but does not flip sign, and the FP16 / BF16 baseline PPL differ by only 0.019 (well below the 0.1 threshold we chose as a proxy for "real fp16-inflation"). We therefore rule out the H1 hypothesis.

Phase 21 (`results/phase21_gemma_b_anomaly_probe.json`) tests reproducibility across two non-overlapping WikiText-2 segments at 30 windows/segment (3× the Phase 19 sample size), and splits (B) into K-only / V-only / full variants. Reproducibility is confirmed: both segments show negative ΔPPL under (B) (-0.40 on segment 1, -0.15 on segment 2). The split confirms K-dominance: (B-K-only) gives 87 % of the B-full effect on both segments; (B-V-only) is small and of inconsistent sign. The data supports the attention-logit entropy-regularization hypothesis more than it supports the value-noise hypothesis, but we do not directly measure entropy and therefore stop short of a mechanistic claim.

Appendix D: Implementation listings

D.1 `qa_nsep_pchan_asym` (the v0.3 recipe)

See §3.5 for the 10-line reference implementation. The packed cache class `Int3AsymNsepPchanCache` is a thin wrapper that stores `packed_d` (uint8 tensor of bit-packed unsigned 3-bit indices), `norm`, `d_min`, and `d_step` on the same device as the input.

D.2 `pack_int3_unsigned` / `unpack_int3_unsigned`

Three-bit unsigned packing: 8 consecutive 3-bit indices in $\{0, \ldots, 7\}$ fit into 3 bytes (24 bits) via a simple bit-shift pattern. The pack is bit-exact; unpack-then-pack is identity on any valid input.

D.3 Quantizer primitives used in the ablation

The four ablation methods use the following quantizer primitives, each exactly matching the mathematical description in §3 and §5.2:

```
(A)  qa_naive_perrow_sym(x, 3)      — per-row symmetric absmax INT3
(B)  qa_asym_pchan_no_nsep(x, 3)    — per-channel [min, max], no norm separation
(C)  qa_nsep_pchan(x, 3)            — symmetric nsep+pchan (v0.2 baseline)
(D)  qa_nsep_pchan_asym(x, 3)       — the v0.3 default
```

All four are implemented in ≤ 20 lines of PyTorch each and operate on the same input shape `[..., T, D]`.

Appendix E: Per-token L2 norm dispersion diagnostic (Phase 23)

Phase 23 (`results/phase23_norm_dispersion.json`) tests the §5.3 mechanism claim (that per-token L2 norm variance contaminates per-channel statistics under variant (B)) by measuring the coefficient of variation of per-token L2 norms on each model's Layer-0 K and V caches.

E.1 Protocol

For each of the eight benchmark models:

1. Tokenize the WikiText-2 test text using the model's own tokenizer. (An earlier character-based slicing produced < 1024 tokens on 7 of 8 tokenizers and was replaced with token-index slicing to guarantee a fixed 1024-token window on every model.)
2. Take three non-overlapping 1024-token segments starting at token indices $\{0, 8192, 16384\}$. All eight tokenized test texts exceed 280 000 tokens, leaving comfortable margin for the third window.
3. For each segment, run a single forward pass with `use_cache=True`, extract the populated `past_key_values` cache.
4. On **Layer 0** (defined as the first *standard-attention* layer, which coincides with cache-index 0 on all dense models and is also cache-index 0 on the hybrid Qwen3.5-9B in our runs) extract the K and V tensors of shape $[1, H, T, D]$ and:
   - pool the $H \times T$ row vectors, compute per-row L2 norms, and report mean, standard deviation, CV, 1st/50th/99th percentiles, and the p99/p50 ratio as a heavy-tail-robust companion to CV;
   - also record per-head CV for reproducibility;
   - save the raw $[H, T]$ norm array for offline histogram plotting.
5. Also compute K-CV at L-mid (cache-index $\lfloor N_{\text{KV}}/2 \rfloor$) and L-last (cache-index $N_{\text{KV}} - 1$) as a coarse depth probe.
6. Report mean ± SD across the three segments for every per-model statistic.

E.2 Why Layer 0

Phase 10 (Appendix A.1) identified Layer-0 K as the locus of the Qwen2-7B outlier structure that motivated recipe development, and Phase 21 (Appendix C) attributed 87 % of the Gemma-4-E4B B-anomaly to K-cache (at unspecified layers). We therefore treat L0 as the primary diagnostic site. The pattern at L0 is not a universal proxy for deeper layers (L-mid and L-last K-CV are lower than L0 K-CV on most models in our set (Table E1)) but L0 is where §5.3's mechanism claim operates, and the L0 V-CV rank order is what matches the variant-(B) failure rank order.

E.3 Full per-layer CV measurements

Table E1. Per-model per-layer CV. L0 rows reproduce Table 2b of §5.3; L-mid and L-last K-CV are included as depth companions. Pythia-6.9B's unusually flat K-CV profile (0.250 → 0.243 → 0.102 across L0 / L-mid / L-last) is distinctive but does not alter the (B) outcome ranking; Qwen2-7B's L-last K-CV of 1.602 is an outlier we do not analyze in this paper and flag for future work.

Model	L0 K-CV	L-mid K-CV	L-last K-CV	L0 V-CV	L0 K p99/p50
Llama-3.1-8B	0.210	0.112	0.158	0.560	1.612
Mistral-7B-v0.1	0.345	0.115	0.150	0.528	1.832
Qwen2.5-14B	0.134	0.110	0.138	0.602	1.172
GPT-J-6B	0.254	0.108	0.124	0.335	1.433
Qwen2-7B	0.472	0.104	1.602	0.472	1.833
Pythia-6.9B	0.250	0.243	0.102	0.347	1.742
Qwen3.5-9B	0.034	0.022	0.024	0.242	1.061
Gemma-4-E4B	0.000	0.000	0.000	0.000	1.000

E.4 Rank correlation summary

Across the 8 models, the Spearman rank correlation between L0 V-CV and variant-(B) regression magnitude (as % of FP16 baseline PPL) is $\rho_V \approx 0.881$; between L0 K-CV and the same is $\rho_K \approx 0.286$. The V-CV / B-failure rank match is exact for 6 of the 8 models and off-by-one for the remaining 2 (GPT-J-6B and Qwen2-7B swap positions 3 and 5 in the V-CV vs B-failure ranking). The K-CV rank is substantially worse because Qwen2-7B (which has the highest L0 K-CV in the set at 0.472) does *not* fail on variant (B), and Llama-3.1-8B / Qwen2.5-14B (whose L0 K-CV is 0.210 and 0.134 respectively) fail catastrophically.

Confidence intervals (Fisher z-transform). With $n = 8$, the Fisher-z 95 % confidence interval for $\rho_V \approx 0.881$ is approximately $[0.46, 0.98]$, and for $\rho_K \approx 0.286$ is approximately $[-0.52, 0.82]$. Both intervals are wide at this sample size, and the K-CV interval in particular crosses zero — our measurement therefore does not rule out the null hypothesis that L0 K-CV is uncorrelated with variant-(B) failure. The conclusion we draw is the *relative* rank of the two correlations (V clearly dominates K in predictive ranking), not a precise estimate of either coefficient.

Confounding. The three catastrophic B-failures in our benchmark cluster on a specific architectural profile ($D = 128$, $H_{\text{KV}} = 8$, GQA, SwiGLU MLPs, Llama-style RoPE; Llama-3.1-8B, Mistral-7B-v0.1, Qwen2.5-14B). At $n = 8$ we cannot distinguish a causal V-CV mechanism from a scenario in which V-CV is merely a proxy for another architectural variable that co-varies with this family — the correlation is therefore *consistent* with the mechanism in §5.3 without confirming it. A larger model set that breaks the family clustering, or a controlled-synthetic probe that perturbs V-CV without touching the rest of the architecture, would separate the two readings; we treat this as primary future work for the mechanism claim.

E.5 Zero-CV observations and QK-normalization

Gemma-4-E4B's L0 K, L0 V, L-mid K, and L-last K all measure CV to within numerical precision of zero, and the corresponding p99/p50 ratio is exactly 1.000 — per-token norms are effectively constant. Qwen3.5-9B's L0 K-CV of 0.034 is almost but not quite zero. We interpret the Qwen3.5 figure as the signature of **architectural QK-normalization**: applying RMSNorm (or equivalent) to Q and K *before* the attention dot product forces K-cache rows to approximately unit L2 norm by construction, which pins per-token K-norm variance at a tight floor instead of letting it grow with model scale. The Gemma-4 K-CV of 0.000 is a stronger version of the same signal. On these two 2026 models, the role of `nsep` on K is redundant with the model architecture; on the six older models it is not. Gemma-4-E4B's L0 V-CV of 0.000 is more unusual (V-side normalization is not standard architectural practice) and we do not characterize it mechanistically in this paper; we flag it as an observation worth confirming against Gemma-4's public architectural documentation when it becomes available.

E.6 Raw data

The JSON record `results/phase23_norm_dispersion.json` contains, for each of the 8 models and each of the 3 segments, per-layer K/V CV, per-head K/V CV, per-token L2 norm percentiles, and the raw $[H, T]$ per-token norm array at Layer 0. This is sufficient to regenerate Table 2b / Table E1, compute the Spearman correlations reported in §5.3 and §E.4, and produce per-model histogram plots without re-running the models.

Appendix F: Multi-needle NIH retrieval probe (Phases 24b / 24c / 24d)

Phases 24b, 24c, and 24d (`results/phase24b_nih_multineedle.json`, `phase24c_nih_multineedle_qwen2.json`, `phase24d_nih_multineedle_qwen25_14b.json`) test whether the proposed recipe preserves long-range factual recall at deployment-scale contexts. A first-pass single-needle probe (Phase 24, `results/phase24_nih_longctx.json`) saturated at 100 % pass rate across all 60 cells on Llama-3.1-8B; the saturated result rules out catastrophic long-context retrieval failure under either INT3 variant but provides no discrimination between methods. The multi-needle distractor-hardened protocol in Phases 24b–24d is the version reported in §5.7.

F.1 Protocol detail

The 10-trial table per (model, context, method) cell is fixed across all three phases for reproducibility. Let `perm` denote the assignment of the three needles to the three haystack positions $\{0.2, 0.5, 0.8\}$ and `target_idx` denote which needle the question asks about:

Trial	`perm`	`target_idx`	Target label	Target is at haystack position
1	$[0, 1, 2]$	0	Titan	0.2
2	$[0, 1, 2]$	1	Orion	0.5
3	$[0, 1, 2]$	2	Vega	0.8
4	$[2, 0, 1]$	0	Titan	0.5
5	$[2, 0, 1]$	1	Orion	0.8
6	$[2, 0, 1]$	2	Vega	0.2
7	$[1, 2, 0]$	0	Titan	0.8
8	$[1, 2, 0]$	1	Orion	0.2
9	$[1, 2, 0]$	2	Vega	0.5
10	$[0, 2, 1]$	0	Titan	0.2 (distractor shuffle: Vega at 0.5, Orion at 0.8)

Trials 1–9 cover every (target label × target position) pair; trial 10 adds one distractor-shuffle case. Each trial is independent — the haystack is rebuilt fresh for each trial, and the model's KV cache is re-populated from scratch. Scoring (pass / wrong-code / hallucinate) is applied to the first 15 greedy-decode tokens after the question's "Answer:" suffix.

F.2 Pooled 32 K outcome

The pooled count at 32 K across the three models (Llama-3.1-8B + Qwen2-7B + Qwen2.5-14B = 30 trials per method) is:

Method	Pass	Wrong-code	Hallucinate	Acc.	vs FP16	vs asym
FP16 baseline	29 / 30	1 / 30	0 / 30	96.7 %	—	$p = 0.194$ (n.s.)

Method	Pass	Wrong-code	Hallucinate	Acc.	vs FP16	vs asym
(C) sym INT3 (v0.2)	15 / 30	6 / 30	9 / 30	50.0 %	$p = 6.4 \times 10^{-5}$	$p = 0.013$
(D) asym INT3 (v0.3)	**25 / 30**	**5 / 30**	**0 / 30**	**83.3 %**	$p = 0.194$	—

All p-values are Fisher exact two-sided. Under this pooled protocol, (D) at 32 K shows no significant difference from FP16 (but $n = 30$ is too small to formally exclude moderate loss); (C) is significantly worse than both FP16 and (D); and the direction of the effect (asym $>$ sym) replicates on all three individual models.

F.3 Complete hallucination list at 32 K

Every hallucinate / wrong-code response under (C) sym INT3 at 32 K, across all three models (15 such responses in total — 2 on Llama, 7 on Qwen2-7B, 4 halluc on Qwen2.5-14B, plus 2 wrong-code not already in the list below). The 9 hallucinations that fuse characters across needles or introduce characters not present in any needle are marked in bold:

Model	target → decoded
Llama-3.1-8B	Titan → `"PQZ5-AMBER-1776"` (wrong-code)
Llama-3.1-8B	Titan → **`"3X-RED-1984"`** (halluc; "RED" absent)
Llama-3.1-8B	Titan → **`"XRZ5-BLUE"`** (halluc; X + Z fusion)
Qwen2-7B	Titan → `"PQZ5-AMBER-1776"` (wrong-code)
Qwen2-7B	Orion → `"XRQ7-BLUE"` (wrong-code)
Qwen2-7B	Vega → `"XRQ7-BLUE-"` (wrong-code)
Qwen2-7B	Titan → `"PQZ5AMBER"` (wrong-code)
Qwen2-7B	Orion → `"XRQ7-BLUE"` (wrong-code)
Qwen2-7B	Vega → `"LMN3-GREEN-19-1776"` (wrong-code)
Qwen2-7B	Orion → **`"XZQ7-RED-"`** (halluc)
Qwen2-7B	Vega → `"XRQ7-RED-19AZ"` (wrong-code)
Qwen2-7B	Titan → **`"LMQ3-GREEN"`** (halluc; L+Q fusion)
Qwen2-7B	Titan → **`"LMQ7-BLUE"`** (halluc; L+Q fusion)
Qwen2.5-14B	Titan → **`"WZQ7-BLUE"`** (halluc)
Qwen2.5-14B	Titan → **`"RQZ3-RED-"`** (halluc)
Qwen2.5-14B	Vega → **`"PQZ7-AMBER"`** (halluc; digit near-miss)
Qwen2.5-14B	Titan → **`"JQZ5-ORAN"`** (halluc; "ORAN" ≈ "ORANGE")

Under (D) asymmetric INT3 at 32 K, the corresponding list has **zero** hallucinate responses and 5 wrong-code responses (e.g., Titan → `"LMN3-GREEN-2037"` and Titan → `"LMN3-GREEN-1984"` on Llama and Qwen2.5-14B; Vega → `"LMN3-GREEN-1776"` on Qwen2.5-14B). The wrong-codes under (D) retrieve a distractor needle's code *verbatim* (not a fused-character synthesis) which is consistent with the mechanism in §5.3 (sym lets per-token V error compound; asym keeps V reconstruction precise enough that, when a retrieval mistake happens, the model still copies one of the three real codes rather than interpolating characters across them).

F.4 The 8 K Qwen2-7B anomaly

At 8 K on Qwen2-7B alone, (D) asymmetric INT3 underperforms (C) symmetric (4 / 10 pass vs 5 / 10 pass). The (D) failures include five hallucinate responses, all of which are within-needle digit / character corruption rather than cross-needle fusion:

target → decoded	Comment
Titan → `"LMQ7-GREEN"`	fuses Titan (Q, 7) with Orion (LM, GREEN)
Orion → `"PQZ3-GREEN"`	fuses Orion (3, GREEN) with Vega (PQZ)
Orion → `"LMN5-GREEN-2037"`	**single-digit corruption** (5 for 3, rest correct)
Titan → `"XRN3-BLUE-2019"`	fuses Titan (XR, BLUE) with Orion (N3); year invented
Titan → `"XRQ5-BLUE-1770"`	fuses Titan (XRQ, BLUE) with Vega (5); year near Vega's 1776

This single-cell anomaly does *not* appear on Qwen2-7B at 32 K, where (D) 7 / 10 cleanly beats (C) 3 / 10, nor on Qwen2.5-14B at 8 K, where (D) 10 / 10 is perfect. We attribute it tentatively to an interaction between Mode-1 K-cache outlier structure (Qwen2-7B's L0 K-CV of 0.472 is the highest in our 8-model set) and per-channel (m, M) estimation variance at short context (fewer tokens to stabilize the asymmetric window); full characterization would require controlled probes at several intermediate context lengths and is left to future work.

F.5 What this probe cannot settle

- It is a retrieval task on a structured synthetic needle format. Natural-language long-context retrieval (e.g., multi-document question answering) may have different failure modes.
- Each cell has only 10 trials; pooling across three models gives $n = 30$, which reaches Fisher-exact significance for the 32 K (C)-vs-(D) comparison but is still small for per-position-slot or per-model sub-analyses.
- Only 8 K and 32 K are tested; the context-scaling curve between and beyond these points is not characterized.
- Greedy decoding with argmax next-token gives deterministic outputs; sampling with temperature could reveal a different noise / retrieval regime we did not probe.